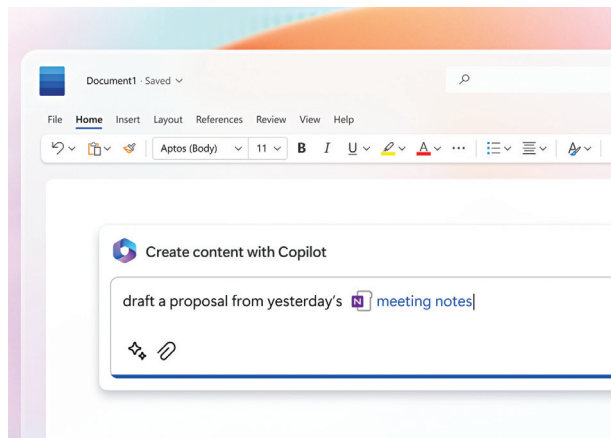


tions like Word, Excel, and PowerPoint. This feature empowered professionals to create documents and presentations efficiently by combining modalities. For instance, a user could dictate a report's introduction, upload charts via visual input, and refine formatting through textual instructions.



The integration of image generation in PowerPoint allowed users to describe a visual concept verbally, prompting the AI to generate relevant slides automatically. This eliminated the need for external design tools, streamlining workflows. Furthermore, by leveraging Azure's AI capabilities, Copilot's multimodal features extended beyond Office, enhancing collaborative platforms like Teams with real-time transcription and visual summarization of discussions.

Apple Vision Pro and AI

Apple's Vision Pro headset in 2024 represented a convergence of multimodal AI and augmented reality, profoundly impacting industries such as design, education, and healthcare. For instance, architects benefited from the ability to manipulate 3D building models with gestures while providing spoken input for specifications. In education, students used eye-tracking to navigate immersive virtual labs, enabling hands-on learning in subjects like chemistry and engineering. Healthcare professionals leveraged the headset for augmented reality-assisted surgeries, combining gesture-based control with verbal guidance. These advancements underscored the headset's ability to bridge virtual and real-world applications, setting new benchmarks for immersive technology. By integrating gesture recognition, voice commands, and eye-tracking, Vision Pro offered an immersive interface where users could interact with both virtual and real-world objects. For example, users could manipulate 3D models with hand gestures, dictate annotations, and navigate menus through gaze.

This system utilized Apple's Neural Engine to process multimodal inputs in real time, ensuring seamless interactions. The device's ability to overlay digital content on physical environments also enabled unique applications, such as virtual

workspaces where users could arrange digital documents using gestures while speaking commands to organize files.

Amazon Alexa with Visual Search

Amazon enhanced Alexa's capabilities by introducing visual search, allowing users to identify objects by pointing a camera. For instance, scanning a book cover with a smartphone would prompt Alexa to provide details about the author, reviews, and purchase options. This feature complemented Alexa's verbal query system, creating a hands-free, multimodal interaction.

In smart home settings, Alexa's multimodal functionality extended to appliance control. Users could visually identify devices through a smartphone's camera, with Alexa providing setup guidance via voice. This dual-modal approach reduced onboarding time for new smart devices, enhancing user accessibility.

Meta AI Research

Meta's advancements in multimodal AI focused on personalizing social media experiences, particularly for content creators and social media users who rely heavily on engagement metrics. For example, Meta's AI models combined text and image analysis to recommend content tailored to individual preferences, increasing user retention by an average of 23%, according to internal studies. In 2024, users uploading photos with captions received automated hashtags and suggestions for enhancing image composition, with over 78% of surveyed



users reporting improvements in audience reach. Meta also introduced tools for content creators, enabling multimodal storytelling. Creators could input text and visuals to generate cohesive multimedia posts, which studies showed resulted in a 35% higher engagement rate compared to traditional posts. This integration highlighted Meta's commitment to blending creative expression with AI-driven personalization, benefiting marketers, influencers, and casual users alike. In 2024, Meta's AI models combined text and image analysis to recommend content tailored to individual preferences. For example, users